

FEDERAL BOARD OF SECONDARY EDUCATION
CHEMISTRY PAPER 26 | ANSWER KEY (Version A)

Marking rules. Total 38. **New: Unit 11 Part 1 = 27. Revision: Unit 8 = 11.** A missing unit or a missing name costs half a mark, the same in every question; it never cancels correct working. A correct method with one arithmetic slip keeps the method marks. Page numbers are from her own printing, and every answer below was read off her book before this key was written.

SECTION A: ANSWER KEY (10 × 1 = 10)

#	Answer	Why	Source
1	(c) atmosphere	Book: "The envelope of gases and water vapour surrounding the planet Earth is called the atmosphere."	p155
2	(a) 78%	Dry air is approximately 78% nitrogen, 21% oxygen, remainder noble gases and carbon dioxide.	p153 SLO, Figure 11.1 p154
3	(d) 4%	"In the tropical rainforest, the air may contain up to 4% water vapour." 21% is the oxygen figure, put there as a distractor.	p155
4	(b) ground level ozone	The only one not emitted directly: it forms in the air from nitrogen oxides and VOCs under sunlight.	p156, 11.2.6
5	(c) carboxyhaemoglobin	"It combines with hemoglobin to form carboxy hemoglobin, which is unable to carry oxygen."	p156, 11.2.4
6	(a) not enough oxygen	"Incomplete combustion occurs when there is not enough oxygen in the combustion process."	p156, 11.2.4
7	(d) NO and NO ₂	"The important oxides of nitrogen that cause air pollution are nitric oxide (NO) and nitrogen dioxide (NO ₂). Collectively they are represented as NO _x ."	p155
8	(b) negative	"Its sign is negative for exothermic and positive for endothermic reactions."	p131, 8.3
9	(c) 25°C (298 K), 1 atm	"Enthalpy of reaction measured at 25°C (or 298K) and one atmospheric pressure is known as standard enthalpy change."	p131, 8.3
10	(a) lower the reaction rate	"The higher the activation energy, the lower the reaction rate."	p132, 8.5

Answer letters: c a d b c a d b c a. Balance a=3, b=2, c=3, d=2, with no letter repeated in consecutive questions.

SECTION B: ANSWER KEY (3 × 4 = 12)

2. Definitions and composition 4 marks (Notes Quiz 1)

Atmosphere (1): "The envelope of gases and water vapour surrounding the planet Earth is called the atmosphere."
Pollutant (1): "Anything that is in the air, water or soil which has a harmful effect on some part of the environment is called pollutant." **All three of air, water and soil are needed;** naming only air scores ½.
Nitrogen 78% and oxygen 21% (1). Water vapour up to 4% in a tropical rainforest (1).

3. The six pollutants, SO_x and NO_x 4 marks (Notes Quiz 2)

The six (1½, ¼ each): sulphur oxides, nitrogen oxides, carbon monoxide, methane, ground level ozone, particulates. (Figure 11.2)
Sulphur oxides (1¼): SO₂ / SO₃, written SO_x (¼). Source, any one (½): burning fossil fuels in power plants, vehicles, industrial units, power generators, residential heating. Effects, any two (½): powerful irritant absorbed in the respiratory system; aggravates asthma, bronchitis, emphysema; causes acid rain; causes haze.
Nitrogen oxides (1¼): NO and NO₂, written NO_x (¼). Source, any one (½): burning fossil fuels in vehicles, power plants, industrial units, generators; car engines produce a great deal. Effects, any two (½): highly toxic; acid rain; photochemical smog; headache; respiratory problems.

4. Carbon monoxide and ground level ozone 4 marks (Notes Quiz 3)

(i) 1½: By **incomplete combustion**, when there is not enough oxygen in the combustion process (¾). It cannot be detected because it is a **colourless and odourless** gas, so its presence cannot be felt (¾). Both words are needed; one alone scores ½.
(ii) 1½: It combines with haemoglobin to form **carboxyhaemoglobin** (¾, the name is compulsory), which is unable to carry oxygen, so the person loses consciousness, suffocates, and it can cause death (¾).
(iii) 1: Because it is not emitted directly. It is produced by complex reactions between **nitrogen oxides** and **volatile organic compounds (VOC)** under the influence of **sunlight**.

SECTION C: ANSWER KEY (2 × 8 = 16)

5. Air pollutants 8 marks

(i) 4 marks, 1 per complete row, any four of the six. Accept any correct source and effect from the book:

Pollutant	Source	Harmful effect
Sulphur oxides	Burning fossil fuels in power plants, vehicles, industry, generators, home heating	Respiratory irritant, worsens asthma/bronchitis/emphysema; acid rain; haze
Nitrogen oxides	Burning fossil fuels in vehicles, power plants, generators; car engines	Highly toxic; acid rain; photochemical smog; headache; respiratory problems
Carbon dioxide	Burning fossil fuels, deforestation, industrial processes	Climate change and global warming
Carbon monoxide	Incomplete combustion in vehicles, industry, heating with fossil fuels and wood	Toxic; forms carboxyhaemoglobin; suffocation and death
Methane	Decomposing vegetation and waste; wetlands, rice fields, landfills; livestock digestion	Increases global warming and so climate change
Ground level ozone	Formed from NO _x and VOCs under sunlight in industrial and urban areas	Smog

(ii) 2 marks: Incomplete combustion means burning that happens when **there is not enough oxygen** in the combustion process (1). It produces **carbon monoxide** (1), and also particulate matter.
(iii) 2 marks: As the heater keeps burning in a sealed room it **uses up the oxygen**, so combustion becomes more and more incomplete and **more carbon monoxide** is produced (1). Because CO is colourless and odourless nobody notices, and it forms carboxyhaemoglobin so the person suffocates. **Precaution (1):** keep a window or ventilator open so fresh air can enter; do not leave a heater burning in a closed room overnight.
Note for the marker: this part is deliberately open. Any precaution that supplies ventilation earns the mark.

6. [REVISION, Unit 8] Enthalpy and bond energy 8 marks

(i) 2 marks: "The amount of heat or thermal energy evolved or absorbed in a chemical reaction is called enthalpy of reaction" (1). Its sign is **negative for exothermic** and **positive for endothermic** reactions (1, ½ each). Source p131.
(ii) 1 mark: $\Delta H^\circ = \text{Sum of bond dissociation energies of reactants} - \text{Sum of bond energies of products}$. Source p131, section 8.4.

(iii) 3 marks. Full working:
 $\Delta H^\circ = [\text{B.E. of H-H} + \text{B.E. of F-F}] - [2 \times \text{B.E. of H-F}]$ (formula, 1)
 $\Delta H^\circ = [436 + 155] - [2 \times 567]$ (substitution, 1)
 $\Delta H^\circ = 591 - 1134$
 $\Delta H^\circ = -543 \text{ kJ/mol}$ (answer with unit, 1; ½ off if the unit is missing)
This is Concept Assessment Exercise 8.2 from her book, p132, never previously set. The method is Example 8.1 on the same page, which worked $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$ to -11 kJ/mol .

(iv) **1 mark: Exothermic**, because ΔH° is **negative**. Saying exothermic without the reason scores $\frac{1}{2}$.

(v) 1 mark. THIS IS THE ITEM THE WHOLE REVISION SECTION EXISTS FOR.

For $2\text{HF}_{(\text{g})} \rightarrow \text{H}_{2(\text{g})} + \text{F}_{2(\text{g})}$, $\Delta H^\circ = +543 \text{ kJ/mol}$.

Reason: reversing a reaction keeps the **same magnitude** and **changes the sign**. The energy released when those bonds formed must be supplied again to break them, so the reverse reaction is endothermic.

Why this is here: on Chemistry exam-25 her single largest loss was Q.7, where a reaction was reversed and she reused the worked example's value instead of re-deriving the direction. A student who writes -543 here, or who recalculates from scratch and gets confused, has not fixed it. **Award the mark only for +543 with the sign explicitly stated.**

Mark distribution check

New (Unit 11 Part 1): MCQs 1 to 7 = 7, Q.2 = 4, Q.3 = 4, Q.4 = 4, Q.5 = 8. **Total 27 (71%).**

Revision (Unit 8): MCQs 8, 9, 10 = 3, Q.6 = 8. **Total 11 (29%).**

All three Quiz questions from her Unit 11 Part 1 notes appear as Section B, Q.2, Q.3 and Q.4.